Sean Luka Girgis

Observability / Production Reliability Engineer | Dynatrace, Python Automation & AI-Assisted Operations

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior observability, performance, capacity, and production reliability professional with 20+ years of enterprise technology experience across Dynatrace/AppMon, CA APM/Introscope, AppDynamics, BMC TrueSight, Python, Bash/KornShell, telemetry pipelines, incident analysis, RCA-style troubleshooting, operational dashboards, runbooks, and AI-assisted engineering workflows. Strong background turning logs, metrics, traces, alerts, and performance signals into actionable operational insights for enterprise systems. Best aligned to observability/SRE-adjacent roles requiring Dynatrace/APM, incident triage, production troubleshooting, Python/shell automation, stakeholder communication, documentation, and human-reviewed operational automation; deep Azure/Kubernetes/Java SRE ownership, GitHub Actions production ownership, Ansible, and multi-agent platform architecture are positioned carefully as adjacent or active-learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Supported enterprise monitoring, capacity, and reliability workflows across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, AWS-backed reporting layers, and large-scale infrastructure telemetry sources.
- Built Python/Pandas/PySpark pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw performance signals into validated dashboards, reports, forecasting datasets, and executive insights.
- Partnered with infrastructure, application, analytics, and engineering teams to investigate abnormal metrics, capacity pressure, bottleneck risks, and production reporting issues.
- Developed KPI-based dashboards and reporting models that converted utilization, application performance, and infrastructure telemetry into actionable views for technical teams and senior stakeholders.
- Troubleshot production monitoring and data issues by tracing logs, source feeds, transformation logic, metric behavior, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Created runbooks, metric definitions, validation checkpoints, reporting logic, support procedures, and knowledge-sharing documentation to improve repeatability and operational handoff.
- Used Python, SQL, Unix/Linux-style automation patterns, and Git-based workflow discipline to support repeatable reporting, data validation, and troubleshooting workflows.
- Applied forecasting workflows with Prophet and scikit-learn to support proactive capacity planning, bottleneck prediction, and risk mitigation across enterprise infrastructure.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, supporting application performance visibility and operational readiness.

- Supported monitoring workflows by configuring dashboards, analyzing synthetic monitoring behavior, and correlating user-facing symptoms with application performance signals.
- Led the FAST project, mining real-user and synthetic monitoring metrics to surface optimization opportunities adopted by engineering teams.
- Integrated HP Performance Center with Dynatrace to correlate load-test behavior with APM telemetry and improve issue diagnosis.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Led and coordinated CA APM upgrades, monitoring standards, dashboard rollouts, threshold design, alerting practices, and operational reporting adoption across technical teams.
- Designed custom Management Modules, dashboards, alerts, SLA thresholds, and documentation standards used by application, middleware, infrastructure, and operations teams.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual performance-reporting workflows.
- Provided technical training, onboarding support, troubleshooting guidance, and documentation to improve adoption of monitoring standards and operational playbooks.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, regression thresholds, test findings, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Supported performance-focused cutover planning, validation, and troubleshooting for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation and integration patterns for structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces and data feed processes for telecom-scale transaction systems.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and operational housekeeping.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.
- Supported production troubleshooting, release support, and cross-team issue resolution for billing and customer-management systems.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Flagship Projects

HorizonScale — Capacity Forecasting Engine
Built Python/PySpark forecasting workflows to process infrastructure telemetry, prepare forecasting datasets, validate pipeline outputs, and deliver capacity and bottleneck insights.
Technologies: Python, PySpark, Prophet, scikit-learn, BMC TrueSight, Telemetry

FAST Performance Analytics Project
Mined Dynatrace and Gomez Synthetic Monitoring data to identify optimization opportunities, improve transaction visibility, and communicate actionable findings to engineering teams.
Technologies: Dynatrace AppMon, Gomez Synthetic Monitoring, Performance Analytics, Dashboards, APM

Enterprise APM Reporting Automation
Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance telemetry across large CA APM enterprise monitoring environments.
Technologies: CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Monitoring, Automation

AI-Powered Job Search Pipeline
Built a Python automation pipeline with LLM-based scoring, structured JSON artifacts, quality gates, semantic duplicate detection, and human-reviewed workflow outputs.
Technologies: Python, LLM API, JSON, FAISS, Sentence Transformers, Quality Gates